A low-threat practice-room interface for neurodivergent game learning

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Fast multiplayer games can convert practice into threat when social judgment, sensory load, ambiguous damage, and repeated failure are experienced as one continuous demand. This manuscript reports a single-user design study of league.aolabs.io, a public review room for League of Legends practice. The system does not claim clinical efficacy, population generality, or competitive optimization. It translates a user-provided learning arc into a low-threat review surface: two champion pages, source-bounded recording review, archived situation notes, public logs, and a downloadable record. The strongest contribution is a design pattern rather than a gameplay result: make the next repetition easier to start by keeping the active champions visible, putting recording feedback first, and moving older situation material into an archive. The interface treats bots as a practice room, human games as performance, death-with-intent as a valid rep, and recording review as mental practice. This bounded case suggests how personal learning software can support high-repetition play without becoming another source of evaluation pressure.

1 Introduction

Expert performance depends on repeated, structured practice, but practice is only useful when the learner can remain in the task long enough for repetitions to accumulate (*1*). In competitive games, especially player-versus-player games, the immediate source of failure is not only mechanical difficulty. The learner must also manage uncertainty, sensory load, social interpretation, and the felt cost of visible mistakes. Anxiety can impair attentional control (*2*); cognitive load can interfere with learning when too many elements must be processed at once (*3*). A player who is trying to learn a champion, lane, camera, item build, opponent behavior, teammate expectations, and self-regulation at the same time may experience the game as a judgment environment rather than a practice environment.

The system described here began from a simple transfer from music practice: keep the task playable, choose one thing, trust repetitions, and let improvement compound. The same principle was applied to League of Legends. The first rule was not to win lane or outplay an opponent; it was to farm and survive. That rule later compressed into the phrase "next safe minion." As the practice record grew, the problem became less about collecting more advice and more about making advice reviewable without overwhelming the player before queueing.

league.aolabs.io is a public AO Labs page built for that purpose. It is not a Riot Games product, a general League guide, a ranking system, or a coaching dashboard. It is a two-champion review room: Samira has current recording review, while Fizz has a clean empty recording page until Fizz footage exists. The design is source-bounded: it uses the user's supplied chat arc, public notes, local recordings, and local client reads as its corpus, and it avoids claims that would require clinical, population-scale, or competitive outcome evidence.

2 Results

2.1 A long learning arc was reduced to a small review surface

The seed corpus contained a long narrative account of League practice. The arc included lane fear, camera stability, sensory control, champion feel, bot practice, death tolerance, public yapping as pressure regulation, and champion-specific rules for Samira, Caitlyn, and Fizz. The current design result is not a long visible essay, a motivational hero, or a broad champion library. The page exposes only Samira and Fizz at the top, puts recording review immediately after those champion cards, and moves older situation notes into an archive.

This reduction is the main interface result. It makes the review surface compatible with the user's stated constraint: the page should be pretty, clean, non-poster-like, and not overwhelming. The long record remains useful as paper, log, and archive context, but the first screen does not ask the user to process a giant review tray, a generic champion taxonomy, or stale situation cards before finding the current recording feedback.

Table 1. Compression of the source arc into champion-specific review units.

Source burden	Review unit	Interface expression
Enemy lane feels like judgment	Archived situation	Next safe minion remains in the paper record
Damage triggers panic	Mindset situation	Damage is information
Samira has many buttons	Button-order situation	Q before E; W saved for real danger
Low targets bait chase	Chase situation	Low plus reachable, not low plus fog
Champion fit affects repetition	Two champion pages	Samira recordings; Fizz empty until footage exists

2.2 Recordings replaced situations as the primary review unit

The site now organizes the visible page by recordings rather than by abstract lesson cards. Situation notes still exist because they explain the origin of the rules, but they are archived below the current recording review. Selecting Samira shows recording rows with the video, game type, champion, match time, game length, K/D/A, CS, one title, and one compact paragraph. Selecting Fizz shows the same page structure with no fake feedback, because no Fizz footage has been source-confirmed. This format was chosen because the user needed enough detail to recognize what happened in the actual game without reading a giant rule wall or a card taxonomy.

Table 2. Example Samira rules retained in the archived situation record.

Situation prompt	Combined response
The fight feels urgent and E wants to become the first button.	Start with Q or an auto so the fight gives information before the dash commits the body.
The enemy has CC that is not recognized yet.	Stay outside on Q-only, watch one danger spell, use W or sidestep for visible projectiles, and E only after the scary spell is gone.
A low-health enemy runs toward fog or teammates.	Stop at the edge; low only matters when the target is also reachable and cheap to kill.
W feels necessary immediately after dashing in.	Hold W for the real spell; it is a parry for visible danger, not armor against being close.
R works and the excitement makes staying feel automatic.	Treat R as the reward, then leave unless the next target is already low and reachable.
E resets after a takedown.	Read the reset as a new check, not an instruction to dash into tank, tower, fog, or fresh crowd control.
One hit lands and feels like proof that the enemy is competent.	Treat damage as information, step back, stabilize the camera, and choose the next action.
The player is near a team fight, R is not ready, and waiting there costs HP.	Move to the edge, build style from safety, and enter only when S rank exists and the fight is already committed.
The player tunnels on one low target.	Check target, danger, target, danger; stop chasing if the chaser becomes the low target.

2.3 Champion focus was narrowed to Samira and Fizz

The visible champion set is no longer a broad library or a tier list. It is deliberately limited to Samira and Fizz. Samira is currently the active focus because she provides a high-reward dash-reset-R loop, but she also creates the central panic trap of using E as an anxiety button. Fizz remains the second page because his button shape makes the desired loop clear: mark, enter, stab, dodge, leave.

The older champion-fit material remains useful as source history, but it is not allowed to compete with the current practice page. The public page uses the two supplied character images as the selection mechanism, keeps Samira recording feedback first, and lets Fizz stay empty until the recording folder contains Fizz footage. This avoids fake specificity and preserves the rule that no footage means no feedback.

2.4 Raw recordings converted a carry game into one improvement target

On May 18, 2026, the public page added a generated recordings manifest from the local League of Legends Highlights folder. The current manifest contains 15 recordings across matches `NA1-5563247854`, `NA1-5563301586`, and `NA1-5563352800`. The set totals 21:21 of video and includes short highlight clips plus three longer full-review recordings. Champion detection was bounded to visible replay evidence: sampled frames show Samira in the team list and center the Soul Fighter Samira model and nameplate. No other player champion was treated as source-confirmed for this recording set.

The recording review does not attempt to become a full coaching dashboard. The public surface now keeps champion cards first, then gives one Samira focus, then lists larger playable recordings with information on the left and video on the right. Each row gives a compact pipe-separated line for queue type, champion, match time, game length, K/D/A, and CS, followed by one title, one paragraph, and the playable video. Rows are sorted with the newest match first and, within a match, the longest recording first. The paragraph is intentionally bounded: it names what happened in the recording, the one improvement, and why that improvement has high-elo transfer value. After manual storyboard review of the newest full recording, the current Samira target is “name the payout before the dash”: before E/R, the player names wave, tower, dragon, recall, or nexus; if none is real, the dash waits.

2.5 The public log made mindset part of the artifact

The public log starts with source entries for the seed arc import, the practice-room frame, the current Samira law, and later user-supplied Samira proof-game notes. This is intentionally public. The design choice is that thoughts, panic notes, and mindset rules are not private by default; they are part of the learning artifact. The only excluded material is unrelated sensitive information such as credentials, access tokens, or private third-party identifiers.

3 Discussion

The case suggests that the useful interface for some high-repetition learners is not less information in the abstract, but better placement of information. A long learning conversation can contain many correct rules while still being hard to use before a game. The revised review room limits the first screen to Samira and Fizz, puts playable recordings before archived situation notes, and gives more detail only where footage exists. That is consistent with cognitive-load theory: reducing simultaneous elements can make practice more tractable (*3*). It is also compatible with mental practice research, because the page gives the user a concrete game moment to rehearse before performance (*4*).

The work also reframes death and failure. The site does not encourage intentional feeding or fake failure. It uses the rule "choose the fight; death is allowed; winning is intended." That distinction matters. It keeps agency with the learner while preserving the real objective of the game. It also separates practice failure from collapse: a failed engage that tested damage, W timing, or exit timing is a rep; walking in only to die is not.

Several limits are important. First, this is a single-user artifact, not a clinical intervention. The user self-described ADHD/autism traits and supplied a practice narrative, but the site does not diagnose, treat, or generalize to neurodivergent players as a class. Second, the current evidence is artifact-level: source text, distilled rules, local recordings, local League Client match-history reads, a rendered page, and the paper itself. It does not include a controlled comparison or a ranked outcome. Third, the system deliberately avoids public Riot API import in version 1. That keeps the artifact focused on review and self-regulation rather than turning it into a general stat tracker.

4 Materials and Methods

Source corpus

The seed corpus was the user-provided League learning arc pasted into the implementation thread on May 17, 2026. It described the user's own League practice, including early Caitlyn farming goals, Samira drills, bot-ladder practice, camera and sensory adjustments, death exposure, and champion-fit comparisons. Later May 17 and May 18 prompts added Samira crowd-control, W-blocking, poke-lane, bait-recognition, team-fight, bad-team, normal-game, and proof-game refinements. A May 18 local recording folder then supplied 15 files across matches `NA1-5563247854`, `NA1-5563301586`, and `NA1-5563352800`. Local replay files supplied the match-time anchor, League client logs supplied queue id `880`, Riot's published queue table identifies `880` as Co-op vs. AI Beginner Bot games,(*5*) and the local League Client match-history endpoint supplied K/D/A, CS, and game duration when the client was available. The implementation treated these texts, recordings, replay files, local logs, and local client reads as the current source of truth for the public situation notes and recording review.

Interface requirements

The page was constrained by user instructions gathered during planning: one page only, clean and pretty, same AO Labs vibe, no poster hero, no dense dashboard, no redundant helper labels, no overexplaining, public notes by default, and one quiet paper strip. The page therefore uses the shared AO Labs header, warm off-white palette, dark ink text, 8 px cards, restrained borders and shadows, compact fragments, and one visible PDF action.

Rule extraction

Rules were extracted by finding repeated gameplay or mindset triggers in the source corpus and translating each into a one-sentence situation prompt plus one combined response paragraph. The extraction favored recognition over slogan compression: the prompt had to name the exact stressful moment, and the response had to combine action, reason, and practice rep without separate "do," "avoid," or "drill" labels. For example, the Samira section retained the underlying rules "Q first; earn the E," "W is parry, not armor," and "R is reward, not start," but the public card now starts from the situation that makes the rule necessary. The death section retained "choose the fight; death is allowed; winning is intended." The bot-practice section retained the practice ladder and the current-mode cue.

Public artifact implementation

The website is served by a small Node application on Railway. It serves the hand-authored page, generated recording manifest, poster frames, and paper artifacts from the public directory and exposes `/api/logs` for public note reads and writes. Large recording videos are kept out of the Railway upload payload; when a local video file is not present in the deployment image, the server redirects that recording request to a GitHub-hosted media copy for the same repository path. A local sync script scans the source Highlights folder, reads replay and League client logs for match and queue metadata, reads local League Client match history for K/D/A, CS, and game duration when available, copies small `.webm` or `.mp4` files into the public recording directory, compresses large recordings into higher-bitrate deployable `.mp4` assets, extracts poster frames with `ffmpeg`, reuses cached feedback for unchanged files, and refreshes the manifest that the browser loads. A local live-recorder script can also watch for the League game process, capture only the League game window with the mouse cursor by default, prefer low-impact hardware H.264 settings when available, lower the capture process priority, keep every captured segment, fast-forward lower-motion segments, preserve higher-motion segments at normal speed, pause post-game processing if another game starts, and hand the resulting shorter whole-game review clip to the same publish path after the game. The recorder stores game-window video and cursor motion, not raw keyboard text. The manifest records match time, queue type, champion, game duration, K/D/A, CS, recording duration, one takeaway, a compact recording-specific paragraph, evidence basis, and review limit for each item, then sorts newest matches first and longest recordings first within each match. On Railway, persistent storage is provided through the configured volume path, and writes are gated by `LEAGUE_WRITE_TOKEN`. This keeps public notes browser-independent without adding a fake local-only note form or claiming public Riot API integration that is not present in version 1.

Acknowledgements

The source learning arc was supplied by Alan Pham during the implementation conversation. Riot Games was not involved in the design, writing, or deployment of this artifact.

Funding

No external funding supported this work.

Author contributions

Alan N. Pham supplied the practice corpus, product constraints, and acceptance criteria. The manuscript and site were assembled from those source materials in the AO Labs workspace.

Competing interests

The author owns and operates AO Labs. The site is a personal public artifact and is not endorsed by Riot Games.

Data availability

The first public data source is the League practice arc supplied in the implementation thread and the public log exposed through league.aolabs.io. The current public page also exposes the May 18–19, 2026 recording manifest for `NA1-5563247854`, `NA1-5563301586`, `NA1-5563352800`, and the automatically captured `NA1-5563660362` review clip: twelve short highlight files, four full-review files, poster frames, local queue metadata, local client K/D/A and CS when available, per-recording feedback, and one main queue focus. The newest automatic clip is stats-backed from local League Client match history because its captured video was produced before the window-only recorder fix. Public Riot API match history is not imported in this version.

Code availability

The implementation is intended for the `nalalalan/league-app` repository, with the public site, PDF, manuscript source, and log API served at league.aolabs.io.

Additional information

References

1. K. A. Ericsson, R. T. Krampe, C. Tesch-Romer, The role of deliberate practice in the acquisition of expert performance. *Psychological Review* **100**, 363–406 (1993).
2. M. W. Eysenck, N. Derakshan, R. Santos, M. G. Calvo, Anxiety and cognitive performance: attentional control theory. *Emotion* **7**, 336–353 (2007).
3. J. Sweller, Cognitive load during problem solving: effects on learning. *Cognitive Science* **12**, 257–285 (1988).
4. J. E. Driskell, C. Copper, A. Moran, Does mental practice enhance performance? *Journal of Applied Psychology* **79**, 481–492 (1994).
5. Riot Games, *League of Legends queue identifiers* (2026; `https://static.developer.riotgames.com/docs/lol/queues.json`).